MKM-TC-001 — Wind-Field Sensitivity Analysis

Component: Tropical Cyclone Progression and Wind-Field — `MKM-TC-001` **Scope:** How a property's priced **peak wind** responds to the two materially uncertain drivers the model produces — storm **intensity** `V_max` and **track offset** (the eye's closest approach) — evaluated over the model's own parametric radial profile. **Date:** 2026-07-30. **Method:** closed-form, data-free; composes `calibrate_outer_decay_length` and `symmetric_profile` (default `WindFieldParams`) via `models.typhoon.wind_field.sensitivity`.

Anchor storm: `V_max = 50 m/s`, `R_max = 30 km`, `R_outer = 250 km`, gale anchor `v_outer_ref = 17.5 m/s`, `alpha_eye = 0.4`, `p = 1.5` (outer-decay length calibrates to `L = 213 km`).

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1. The two material inputs

The symmetric profile is a linear eyewall ramp inside `R_max` and a stretched- exponential decay outside it:

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inner (r < R_max):  V(r) = V_max * [ alpha_eye + (1 - alpha_eye) * r/R_max ]
outer (r >= R_max): V(r) = V_max * exp( -((r - R_max)/L)^p )
L calibrated so V(R_outer) = v_outer_ref (gale)
```

Peak wind at a property therefore depends on the storm's intensity `V_max` and on where the property sits relative to the eye, `r`. Both are outputs of the Bayesian particle filter and both carry real uncertainty; storm size (`R_max`, `R_outer`) is held at the anchor here.

2. Intensity — near-proportional passthrough

The whole field scales with `V_max`, but the outer length `L` is re-anchored to the gale radius as `V_max` moves, so a property in the outer field grows slightly **sub**-linearly. At an offset of 60 km:

V_max factor	V_max (m/s)	local wind (m/s)	rel. to base
0.8	40	38.4	0.81
1.0	50	47.4	1.00
1.2	60	56.4	1.19

Passthrough to local wind is close to one. The material amplification is downstream: the wind-damage curve (MKM-WD-001) is a steep sigmoid in gust speed about a 50%-damage threshold `v_50`, so this near-proportional wind error becomes a **larger** loss error once it passes through vulnerability. This model's own output — peak wind — is not where intensity error is magnified.

3. Track offset — the dominant geometric uncertainty

The profile is steepest on the eyewall ramp. Local wind climbs from `alpha_eye * V_max = 20 m/s` at the centre to the full `50 m/s` at `R_max` — a factor `1/alpha_eye = 2.5` over 30 km — then decays gently once the gale radius is far out:

offset r (km)	15	30	45	60	90	120
local wind (m/s)	35.0	50.0	49.1	47.4	43.1	38.0
gradient dV/dr (m/s per km)	+1.00	—	-0.09	-0.13	-0.16	-0.17

The eyewall gradient (about **1 m/s per km**) is roughly **six to ten times** steeper than the outer decay. So whether the eyewall band (`r ~ R_max`) sweeps over a property, versus passing tens of km inside or outside it, swings its peak wind far more than a plausible intensity error does.

There is a subtlety the model gets right: peak wind is **not** monotone in closest approach. Propagating an uncertain offset shows it —

percentile	offset r (km)	local wind (m/s)	move vs median
p05	10	30.0	-19.1
p50	45	49.1	—
p95	90	43.1	-6.0

A track passing **right over the eye** (10 km) gives a *lower* wind (30 m/s) than one clipping the eyewall at 45 km (49 m/s), because the eye is calm. A validation or exposure read that assumed “closer is always worse” would mis-rank exactly the near-miss geometries that dominate the tail.

4. What this does and does not establish

The winds and gradients are exact readings of the calibrated radial profile at the stated anchor, not sampled estimates; they bound how peak wind *responds* to intensity and track error. They do not size those errors — the particle filter’s track and intensity spread is what sets the actual uncertainty, and on synthetic tracks it is unvalidated (validation questions VQ-005 data quality and VQ-009 independent validation — outstanding). The control implication is that track placement near `R_max`, not intensity precision, is the highest-value target for the per-location wind, and that the eye-passage non-monotonicity must be preserved in any downstream exposure ranking.

Reproduce: `models.typhoon.wind_field.sensitivity` — `intensity_sensitivity`, `track_offset_sensitivity`, `peak_wind_distribution`; figures from the default `WindFieldParams` and the stated anchor storm, 2026-07-30.